

Report for Credit Card Fraud Detection Using HE Framework

Introduction

This project focused on implementing a comprehensive workflow for credit card fraud detection using machine learning models integrated with homomorphic encryption (HE) to ensure privacy-preserving computations. By leveraging tools such as Pyfhel for encryption, this approach demonstrated the ability to detect fraudulent transactions efficiently and securely while protecting sensitive data from exposure.

1. Code Overview

The notebook achieves the following objectives:

A. Data Preprocessing:

- The dataset is preprocessed to handle missing values, normalize features, and balance classes.
- StandardScaler is used to standardize the feature set for better model performance.

B. Machine Learning Models:

- **Logistic Regression** and **Random Forest Classifier** models are used to predict fraudulent transactions (isFraud field).
- Evaluation metrics such as ROC AUC, confusion matrix, and classification reports highlight the models' performance.

C. Homomorphic Encryption (HE):

- TenSEAL, a library for HE, is integrated into the workflow.
- Model predictions are encrypted to ensure that sensitive outputs are not exposed.
- Encrypted predictions are decrypted securely, and results are interpreted as "High Risk," "Moderate Risk," or "Low Risk."

D. Risk Analysis:

- The encrypted predictions are analyzed for fraud risk levels using thresholds.
- The workflow ensures privacy while enabling actionable insights.

E. Visualization:

- The ROC curve is plotted to assess the classifier's performance visually.

2. Uses and Applications

1. Fraud Detection:

- The workflow can be integrated into banking and financial systems to detect anomalies in real-time without compromising user privacy.
- Ensures compliance with data protection laws like GDPR by encrypting sensitive data.

2. Privacy-Preserving Analytics:

- By encrypting predictions, this approach prevents sensitive customer information from being exposed during fraud detection.

3. Actionable Insights:

Provides fraud risk categorization to help banks prioritize investigations and take proactive measures.

4. **Scalability:**

- With optimizations for parallelism, the HE framework can handle larger datasets and complex models in the future.

Next Steps

1. **Enhanced Model Performance:**

- Explore advanced models such as Gradient Boosting Machines (e.g., XGBoost) or neural networks to improve prediction accuracy.
- Conduct hyperparameter tuning to optimize model performance.

2. **Integration with Real-World Data:**

- Test the workflow with real-world credit card transaction datasets to validate its robustness.
- Address domain-specific challenges like imbalanced datasets or skewed transaction types.

3. **Scalability Improvements:**

- Implement batching in encryption and decryption workflows to handle larger datasets efficiently.
- Increase the `poly_modulus_degree` in Pyfhel to process more complex computations securely.

4. **User-Friendly Interface:**

- Develop a front-end dashboard for end-users (e.g., banks) to visualize fraud risk levels and predictions.

5. **Extended Use Cases:**

- Adapt the HE-based approach for other domains, such as healthcare, for detecting anomalies in encrypted medical data.

6. **Cloud Integration:**

- Deploy the solution on cloud platforms like AWS or GCP for real-time fraud detection with secure data handling.

Conclusion - This project demonstrates the potential of combining machine learning and homomorphic encryption for secure and efficient fraud detection. By leveraging privacy-preserving techniques, the workflow ensures compliance with stringent data protection regulations while delivering actionable insights. Future work will focus on improving model accuracy, scalability, and usability to deploy the solution in real-world settings. This effort underlines the critical role of encryption technologies in advancing secure analytics for sensitive applications.